Microbiology: Opportunistic Mycoses

NEW YORK INSTITUTE OF TECHNOLOGY

Joseph Holloway, Ph.D., M.Sc.

Department of Biomedical Sciences

Email: jhollo03@nyit.edu



Do.
Make.
Innovate.
Reinvent the Future.

Session Objectives

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

1. List and describe the opportunistic fungal pathogens.
2. Describe the risk factors that lead to opportunistic fungal infections.
3. Describe the diseases caused by opportunistic fungal pathogens
4. Outline the laboratory techniques used for the identification and diagnosis of opportunistic fungal pathogens
5. List the treatment options for the various mycoses

Introduction

- Opportunistic mycoses
- High-risk patients:
  - Immunosuppressed state (e.g. AIDS)
- Other risk factors:
  - Large burn wounds
  - Trauma
  - Diabetes
  - Renal insufficiency requiring dialysis
  - Medical devices

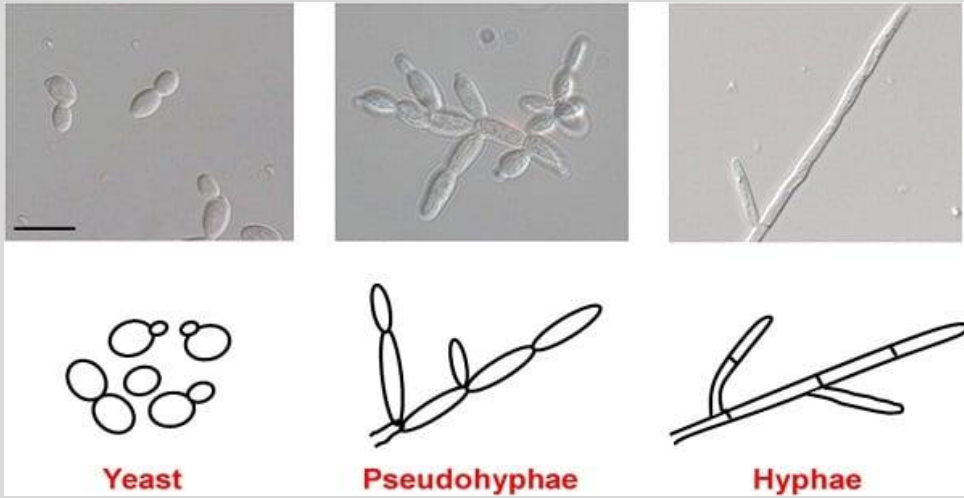
- Opportunistic fungal diseases:
  - Candidiasis
  - Aspergillosis
  - Cryptococcosis
  - Pneumocystis
  - Mucormycosis

Candidiasis

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

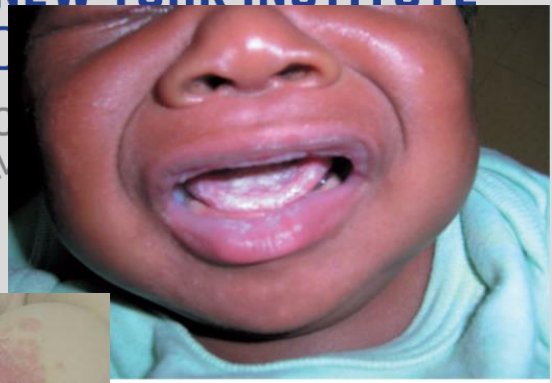
- *Candida* spp. part of normal human flora (endogenous opportunist)
- Most prevalent systemic mycoses
- *C. albicans*
    - Others: *C. glabrata, C. auris*
- Round/oval yeasts form hyphae or pseudohyphae

- Immune defenses to prevent infection:
    - Skin & mucous membranes
    - Cell mediated immunity
    - Neutrophils



Candidiasis: Clinical Manifestations

- **Oral Thrush:**
  - Overgrowth in the mouth
  - Adherent, painless, white plaques or patches



Levinson: Review of Medical Microbiology
, 14th Edition, www.accessmedicine.com
Graw-Hill Education. All rights reserved.

- **Vaginal Yeast Infection:**
  - Normal flora of vaginal tract
  - Can overgrow if ecology disrupted



- **Cutaneous Infection:**
  - Commonly in warm, moist areas of the body (possibly fingers/nails)
  - Diaper rash



Source: K. Wolff, R.A. Johnson, A.P. Saavedra, E.K. Roh:
Fitzpatrick's Color Atlas and Synopsis of Clinical
Dermatology, Eighth Edition: www.accessmedicine.com
Copyright © McGraw-Hill Education. All rights reserved.

Candidiasis: Clinical Manifestations

NEW YORK INSTITUTE OF TECHNOLOGY

- **Chronic Mucocutaneous Candidiasis:**
  - Prolonged infections
  - IL-17 issues predispose

- **Disseminated candidiasis:**
  - Spreads from blood to other areas
  - Caused by:
    - Immunosuppression
    - Medical procedures and devices
    - IV drug use



Source: Usatine RP, Smith MA, Mayeaux EJ, Chumley HS: *The Color Atlas of Family Medicine, Second Edition*: www.accessmedicine.com
Copyright © The McGraw-Hill Companies, Inc. All rights reserved.



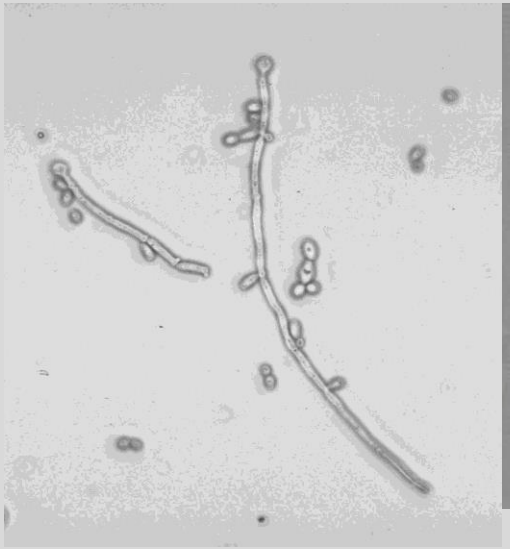
Source: D. L. Kasper, A. S. Fauci, S. L. Hauser, D. L. Longo, J. L. Jameson, J. L
www.accessmedicine.com
Copyright © McGraw-Hill Education. All rights reserved.

Candidiasis

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- **Diagnosis:**
  - Visualize budding yeasts, hyphae or pseudohyphae
  - *Candida albicans* forms germ tubes at 37°C
  - Sugar fermentation for non-albicans spp.
  - Culture
  - PCR

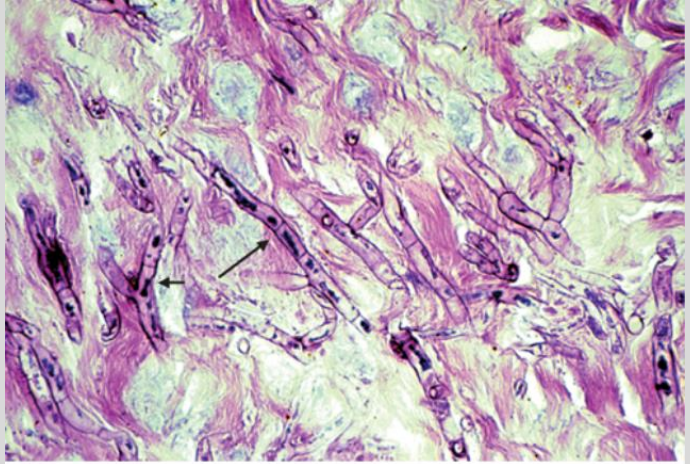
- **Treatment:**
  - Topical nystatin or azole
  - Azoles (vaginal or oral)
  - Amphotericin B with an azole or caspofungin (disseminated)





Aspergillosis

- Common *Aspergillus* species that cause infection
  - *A. fumigatus, A. flavus, A. niger*
- Not part of normal human flora
- Mold only
- Form septate hyphae that form V-shaped (acute angles) branches
- Conidia (spores) form radiating chains



Source: Warren Levinson: Review of Medical Microbiology and Immunology, 14th Edition, www.accessmedicine.com
Copyright © McGraw-Hill Education. All rights reserved.





Aspergillosis

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- **Risk factors:**
  - Immunocompromised state
  - Neutropenia
  - Glucocorticoid use

- **Pathogenesis:**
  - Spores often inhaled into respiratory tract
    - Can grow in cavities within lungs
    - Can disseminate in blood
  - Able to invade abraded skin, burns, cornea, & sinuses

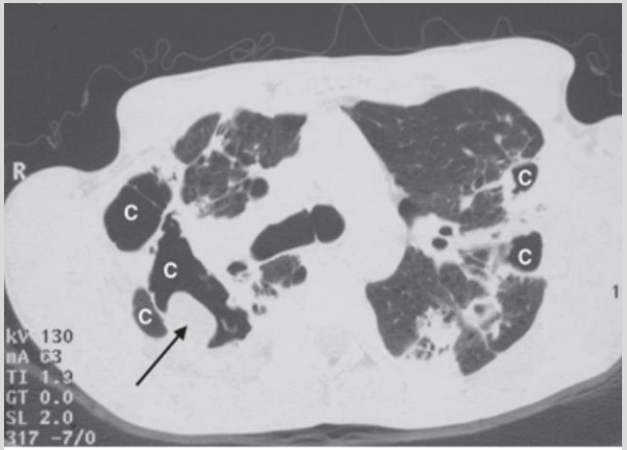


Source: Lichtman MA, Shafer MS, Felgar RE, Wang N: *Lichtman's Atlas of Hematology*: http://www.accessmedicine.com
Copyright © The McGraw-Hill Companies, Inc. All rights reserved.

Aspergillosis: Clinical Manifestations

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- **Invasive Pulmonary Aspergillosis:**
  - Acute
    - Symptoms: fever, cough, chest discomfort, hemoptysis, shortness of breath
    - Often disseminates
  - Chronic
    - Growth in pulmonary cavities over time
    - Can display an aspergilloma (fungal ball)



Source: D. L. Kasper, A. S. Fauci, S. L. Hauser, D. L. Longo, J. L. Jameson, J.
www.accessmedicine.com
Copyright © McGraw-Hill Education. All rights reserved.

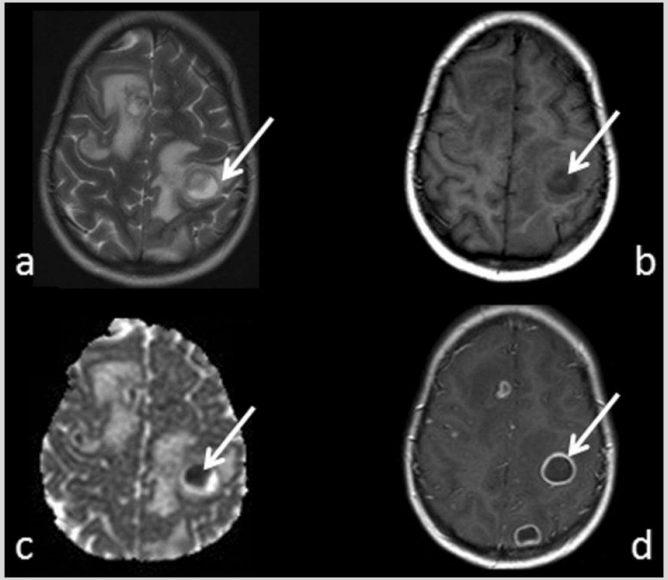
C= cavity, black arrow points to aspergilloma

Aspergillosis: Clinical Manifestations

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- **Disseminated Aspergillosis:**
  - Severely immunocompromised
  - Can spread to brain, skin, thyroid, bone, kidney, liver, GI tract, eye, & heart valves
  - Can form abscesses

- **Allergic Bronchopulmonary Aspergillosis (ABPA):**
  - Hypersensitivity reaction to *A. fumigatus*
  - Bronchial obstruction with mucous plugs, breathlessness
  - Elevated IgE, eosinophilia

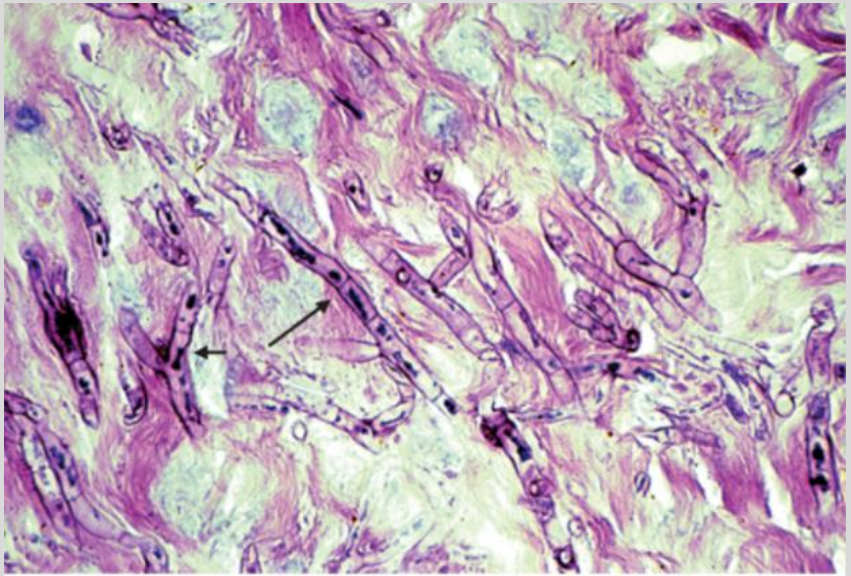


Aspergillosis

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- **Diagnosis:**
  - Microscopic visualization
  - Culture
  - Serology
    - Galactomannan antigen
  - PCR
  - ABPA patients: IgE titer

- **Treatment:**
  - Voriconazole
  - Amphotericin B, posaconazole or caspofungin
  - Surgery

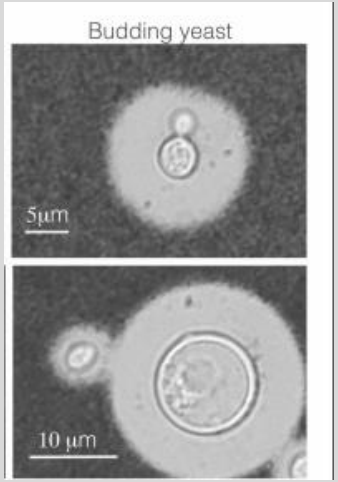


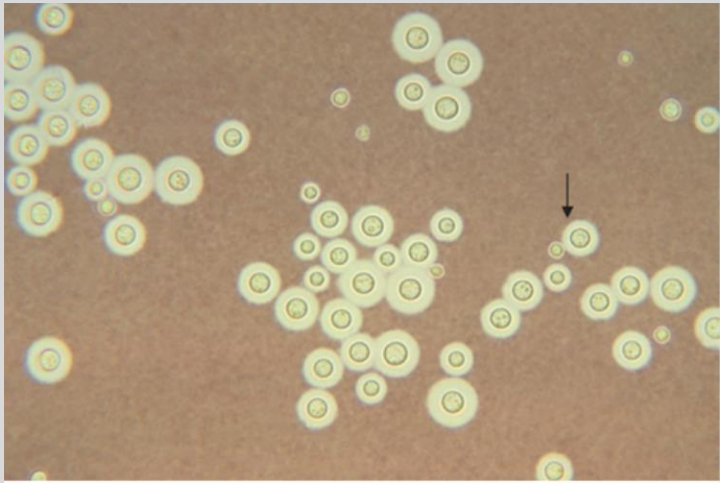
Source: Warren Levinson: Review of Medical Microbiology and Immunology, 14th Edition, www.accessmedicine.com
Copyright © McGraw-Hill Education. All rights reserved.

Cryptococcosis

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- *Cryptococcus* genus
  - *C. neoformans, C. gattii*
  - Four serotypes
- Oval budding yeast
  - Form narrow-based buds
- Large polysaccharide capsule





Source: Warren Levinson: Review of Medical Microbiology and Immunology, 14th Edition, www.accessmedicine.com
Copyright © McGraw-Hill Education. All rights reserved.

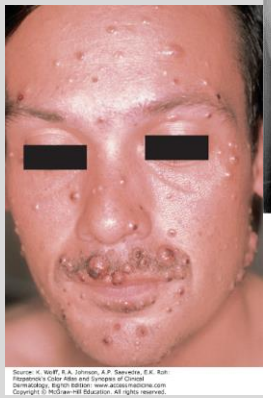
- Risk factors:
  - Immunosuppression
  - Organ transplants
  - Glucocorticoids

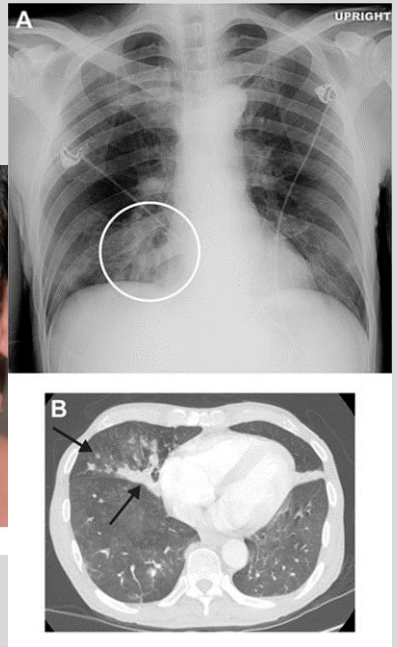
- Naturally found in the GI tract of pigeons

Cryptococcosis: Clinical Manifestations

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- **Pulmonary infection:**
  - Usual site of infection
  - Cough, increased sputum, chest pain, etc.
  - Most cases self resolve
- **Disseminated:**
  - Cutaneous lesions
    - Highly variable: papules, plaques, vesicles, etc.
  - Meningitis
    - Headache, fever, neurological complications
- Can form cryptococcomas (granulomatous masses)

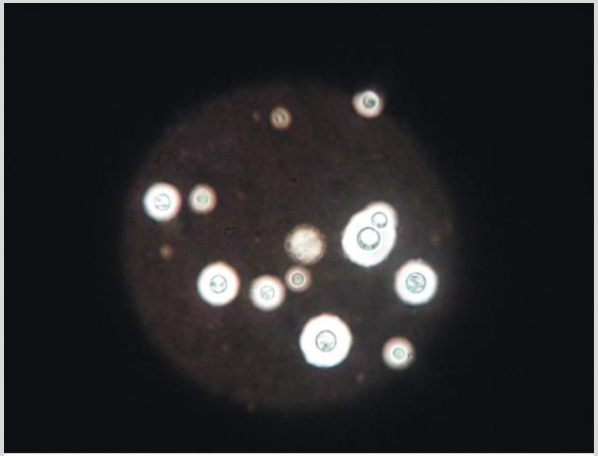




Cryptococcosis

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- **Diagnosis:**
  - Visualization in cerebrospinal fluid (CSF)
    - India ink stain reveals fungal cell with large capsule
  - CSF/blood cultures
  - Cryptococcal capsular antigen (CRAg) testing in CSF & blood -latex agglutination test
  - PCR
- **Treatment:**
  - Amphotericin B & flucytosine (disseminated/meningitis)
  - Fluconazole (long term treatment or lung infections)



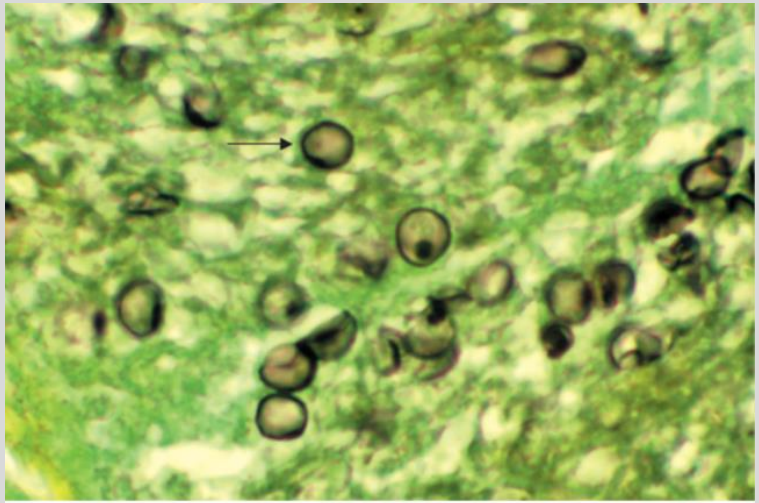
Source: K.J. Knoop, L.B. Stack, A.B. Storrow, R.J. Thurman:
The Atlas of Emergency Medicine, 4th Edition,
www.accessemergencymedicine
Copyright © McGraw-Hill Education. All rights reserved.

Pneumocystis

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- *Pneumocystis jirovecii*
  - (formerly *P. carinii*; thought to be a protozoan)
- Appears as cysts in tissue
- Possibly normal flora

- Risk factors:
  - Immunosuppression (HIV infection)
  - Organ transplants
  - Glucocorticoids
- Defense: CD4+ T cells are critical



Source: Warren Levinson: Review of Medical Microbiology and Immunology, 14th Edition, www.accessmedicine.com
Copyright © McGraw-Hill Education. All rights reserved.

Pneumocystis: Clinical Manifestations

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- **Pneumocystis Pneumonia (PCP):**
  - Presents as acute or subacute pneumonia
  - Dyspnea, fever, non-productive cough
  - High mortality rate if untreated

- Chest X-ray shows “ground glass” pattern

- **Extrapulmonary manifestations**
  - Rare
  - Lymph nodes, spleen, liver



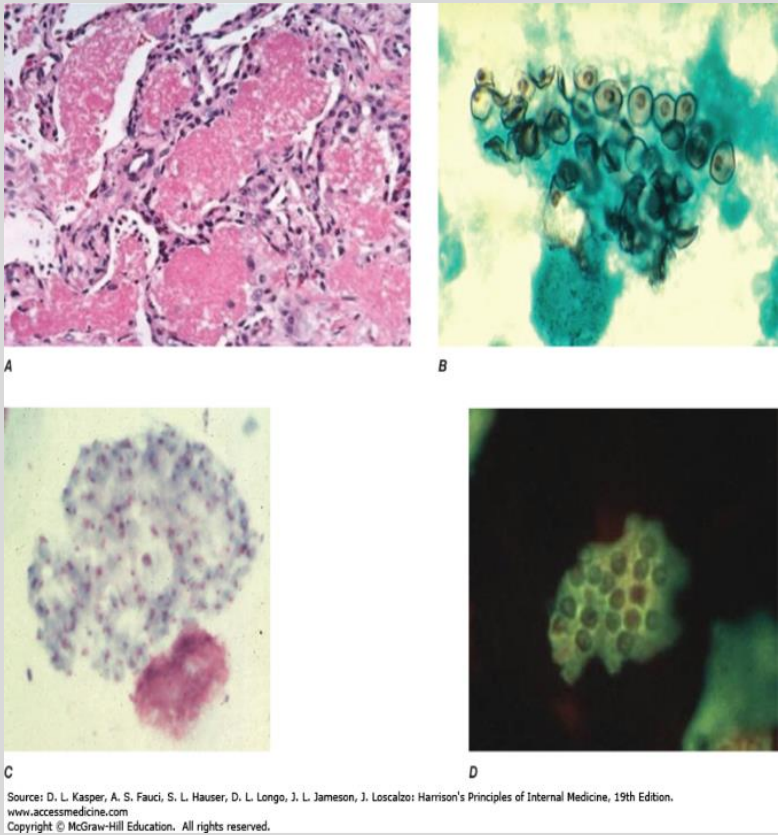
Source: K.J. Knoop, L.B. Stack, A.B. Storrow, R.J. Thurman:
The Atlas of Emergency Medicine, 4th Edition,
www.accessemergencymedicine
Copyright © McGraw-Hill Education. All rights reserved.

Pneumocystis

NEW YORK INSTITUTE OF TECHNOLOGY

- **Diagnosis:**
  - Visualization of organism
    - Fluorescent antibody stains
    - Giemsa, silver stain
  - PCR
  - Is not culturable

- **Treatment:**
  - Trimethoprim-sulfamethoxazole (TMP-SMX)
  - Pentamidine



Source: D. L. Kasper, A. S. Fauci, S. L. Hauser, D. L. Longo, J. L. Jameson, J. Loscalzo: Harrison's Principles of Internal Medicine, 19th Edition.
www.accessmedicine.com
Copyright © McGraw-Hill Education. All rights reserved.

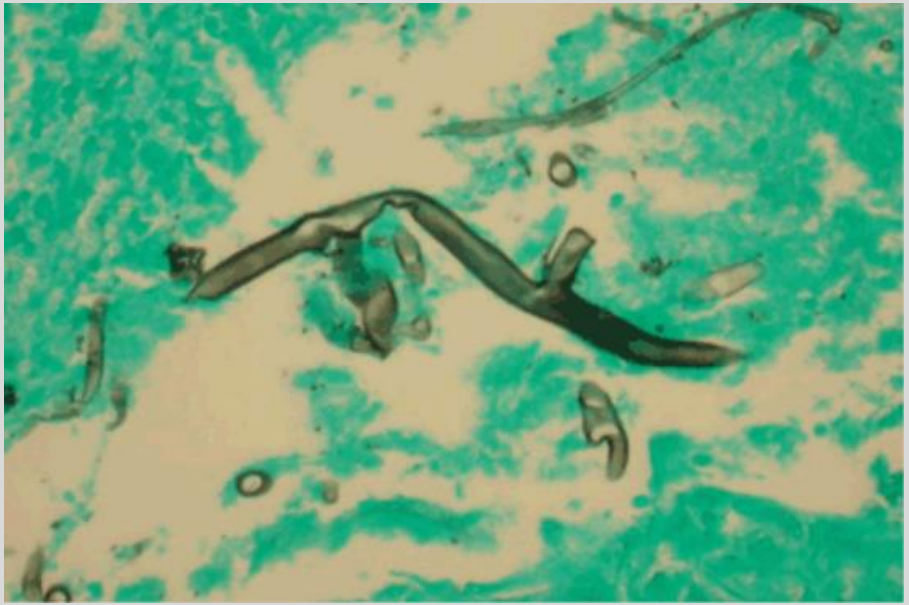
Mucormycosis

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- *Rhizopus & Mucor* Genera
- Formerly called Zygomycosis
- *Rhizopus oryzae* most common

- Broad hyphae, non-septated, irregular branching at right angles

- Risk factors:
    - Immunosuppression
    - Elevated free iron
    - Diabetes (ketoacidosis)



Mucormycosis: Clinical Manifestations

NEW YORK INSTITUTE OF TECHNOLOGY

- **Rhino-Orbital-Cerebral Disease:**
  - Non-specific sinusitis symptoms
  - Spreads to orbit, palate, facial soft tissue
  - Can form black necrotic eschar
- **Pulmonary Disease:**
  - Dyspnea, cough, hemoptysis, chest pain, fever
- **Cutaneous Disease:**
  - Invasive; necrotizing
- **Gastrointestinal Disease**
- **Disseminated Disease**





Source: Warren Levinson: Review of Medical Microbiology and Immunology, 14th Edition, www.accessmedicine.com
Copyright © McGraw-Hill Education. All rights reserved.

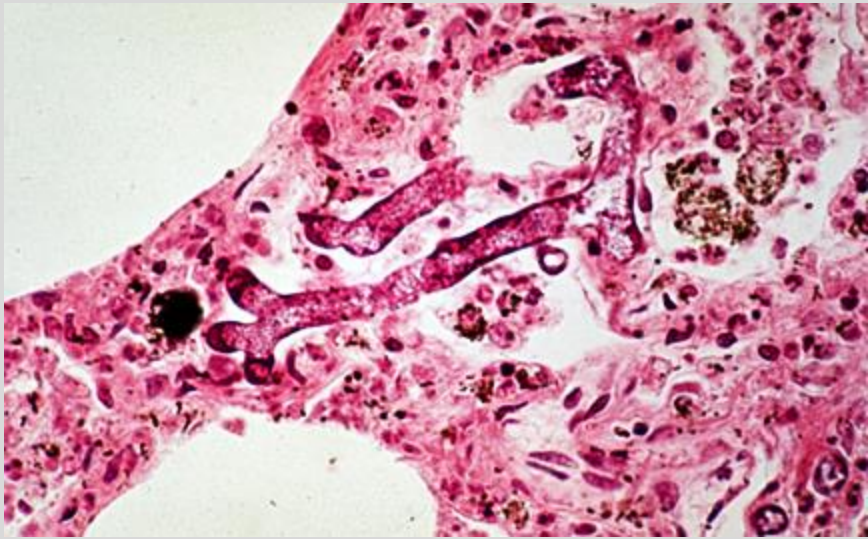




Mucormycosis

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- **Diagnosis:**
  - Visualization of organism
  - PCR
  - Culture is difficult
- **Treatment:**
  - Amphotericin B
  - Surgical debridement



Summary

Disease	Organism (most common)	Clinical Features	Diagnosis	Treatment
Candidiasis	*Candida albicans*	Mucocutaneous Can disseminate to eyes, kidney, heart, brain, liver, spleen, skin	*C. Albicans* germ tubes at 37°, Mucocutaneous=visualize yeasts, hyphae or pseudohyphae on slides, disseminated=culture	Disseminated=amphotericin B, Cutaneous=topical azoles; nystatin
Aspergillosis	*Aspergillus fumigatus*	Invasive Pulmonary Aspergillosis, Sinusitis, ABPA, disseminated	Biopsy shows septate hyphae that branch at 45°, conidia in radiating chains Culture	Voriconazole 1st line, Caspofungin, posaconazole, amphotericin B 2nd line
Cryptococcosis	*Cryptococcus neoformans*	meningitis, pulmonary disease, cutaneous disease	India ink stain reveals fungal cell with large capsule Capsular antigen testing—latex agglutination	Combined Amphotericin B and flucytosine Pulmonary disease: fluconazole
Pneumocystis	*Pneumocystis jirovecii*	Pneumocystis pneumonia, extrapulmonary symptoms rare	Diffuse interstitial pneumonia with "ground glass" infiltrations, small round cysts	trimethoprim-sulfamethoxazole (TMP-SMX) pentamidine
Mucormycosis *diabetes=unique risk factor	*Rhizopus, Mucor species*	Rhino-Orbital-Cerebral Disease, pulmonary, cutaneous, GI, disseminated	Wide, thick-walled, ribbon-like, nonseptate hyphae with branching at right angles	Amphotericin B

*immunocompromised state is a risk factor for all opportunistic mycoses

NEW YORK INSTITUTE
OLOGY
teopathic

References

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- Chin-Hong P. et al. 2026. Levinson's Review of Medical Microbiology & Immunology, 19th ed. McGraw-Hill Education.
- Murray, PR et. al. 2026. Medical Microbiology, 10th ed. Elsevier.
- Tao, LE et. al. 2026. First Aid for the USMLE Step 1; McGraw Hill Publication.
- UptoDate:
- https://nyit.idm.oclc.org/login?url=https://www-uptodate-com.nyit.idm.oclc.org/contents/overview-of-candida-infections
- https://nyit.idm.oclc.org/login?url=https://www-uptodate-com.nyit.idm.oclc.org/contents/epidemiology-and-clinical-manifestations-of-invasive-aspergillosis
- https://nyit.idm.oclc.org/login?url=https://www-uptodate-com.nyit.idm.oclc.org/contents/microbiology-and-epidemiology-of-cryptococcus-neoformans-infection
- https://nyit.idm.oclc.org/login?url=https://www-uptodate-com.nyit.idm.oclc.org/contents/epidemiology-clinical-manifestations-and-diagnosis-of-pneumocystis-pneumonia-in-patients-without-hiv
- https://nyit.idm.oclc.org/login?url=https://www-uptodate-com.nyit.idm.oclc.org/contents/mucormycosis-zygomycosis

THANK YOU!

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R